MATH 127 Calculus for the Sciences

Lecture 31

Today's lecture

Last time
Optimization
linearization

This time
Course note coverage Section 4.2.2, 4.2.3
Binomial Approximation
Quadratic Approximation

Binomial Approximation is just an application of linear approximation to binomials

The expression

$$(1+x)^k, \;\; k \text{ is some integer (could be negative)}$$

is called a **binomial**. We want to approximate $f(x) = (1+x)^k$ for some k, when x is really close to 0.

Recall what tools we have so far:

1. If you want to approximate $f(a) - f(b)$ for some b close to a, you could use $\boxed{}$.
2. If you want to approximate $f(x)$ for some x close to a, you could use $\boxed{}$, aka $\boxed{}$.
3. If you want to approximate $\int_a^b f(x)dx$, you could use $\boxed{}$.

Binomial Approximation

So let's apply linear approximation to $f(x) = (1+x)^k$ at $x = 0$.

$$f(x) \approx \boxed{} = \boxed{}$$

Binomial Approximation

Theorem (Binomial approximation) *When x is very close to 0, we can approximate*

$$(1+x)^k \approx 1+kx$$

k doesn't even have to be an integer, it can be any number!

Theorem (Binomial approximation) *When $\boxed{g(x)}$ is very close to 0, we can approximate*

$$(1+g(x))^k \approx 1+kg(x)$$

$g(x)$ can be anything you want.

Example

Example Approximate

$$\sqrt{3+\ln(x)}$$

when x is really close to 1 using binomial approximation.

1. The function needs to look like $(1+g(x))^k$ for us to apply binomial approximation at all. So we rewrite

$$\sqrt{3+\ln(x)} = \sqrt{3}\cdot\sqrt{1+\frac{\ln x}{3}} = \sqrt{3}\cdot\left(1+\frac{\ln x}{3}\right)^{0.5}$$

2. This means we can set $g(x) = \boxed{}$ and $k = \boxed{}$.
3. You have to state the premise: when x is close to 1, $g(x)$ is very close to $\boxed{}$, so it's valid to apply binomial approximation. If you don't state this, your application is not justified
4. Conclusion: binomial approximation gives us

$$\sqrt{3}\cdot\left(1+\frac{\ln x}{3}\right)^{0.5} \approx \sqrt{3}\cdot\boxed{}$$

Quadratic approximation

Lines, aka linear functions look like

$$y = mx + b.$$

But we prefer the point-slope form

$$\boxed{y = m(x - a) + c}$$

because this tells us the line has slope m and passes through the point (a, c).

Using this we got tangent line approximations, aka linear approximations.

What's better than lines? Parabola! This is a hot take, don't quote me on this.

Quadratic approximation

Parabolas, aka qudratic functions look like

$$y = Ax^2 + Bx + C.$$

But we prefer the “point-derivative" form

$$\boxed{y = \frac{m_2}{2}(x-a)^2 + m_1(x-a) + c}$$

because this tells us the parabola (a, c), has first derivative m_1 at $x = a$ and second derivative m_2 at $x = a$.

So can we approximate functions using parabolas just like what we did with lines, like some kind of “tangent parabola approximation"?

Yes, this is called **quadratic approximation**.

Quadratic approximation

The **linear approximation** for x near a is

$$f(x) \approx P_{1,a}(x) = f'(a)(x-a) + f(a)$$

The **quadratic approximation** for x near a is

$$f(x) \approx P_{2,a}(x) = \frac{f''(a)}{2}(x-a)^2 + f'(a)(x-a) + f(a)$$

Remark

1. $P_{1,a}(x)$ is called the Taylor polynomial of $f(x)$ at a of degree 1.
2. $P_{2,a}(x)$ is called the Taylor polynomial of $f(x)$ at a of degree 2.

Because we are using one more degree than before, in general, we know quadratic approximation is more accurate than linear approximations. The downside is it's harder to compute, because it's more complicated.

Quadratic approximation

Example

1. What is the degree 2 Taylor polynomial of e^x at $x = 0$?
2. Approximate $e^{0.05}$ using quadratic approximation.

1. By the formula from the previous slide, we have

$$P_{2,0}(x) = \frac{f''(0)}{2}(x-0)^2 + f'(0)(x-0) + f(0) = \boxed{\quad\quad\quad\quad}$$

2. By the formula from the previous slide, we approximate

$$e^{0.05} \approx P_{2,0}(0.05) = \boxed{\quad\quad\quad\quad} = \boxed{\quad\quad}$$

Remark The true value of $e^{0.05}$ is 1.051271....

Exercise

Challenge Recall that binomial approximation is just applying linear approximation to the binomial $(1+x)^k$.

What happens if you apply quadratic approximation instead? You would get a more accurate approximation. Calculate what that would be, and make up a name for it.

Kelly's Criterion

Let me show you an example of optimization in the field of finance/investment/gambling.